7/3/23

WN: $W_t \sim WN(0, \sigma^2_\omega)$

1. $\{W_t\}$ is a time series
2. $E(W_t) = 0$
3. $V(W_t) = \sigma^2_\omega$ fixed
4. $corr(W_t, W_s) = 0 \quad \forall\, t \neq s$

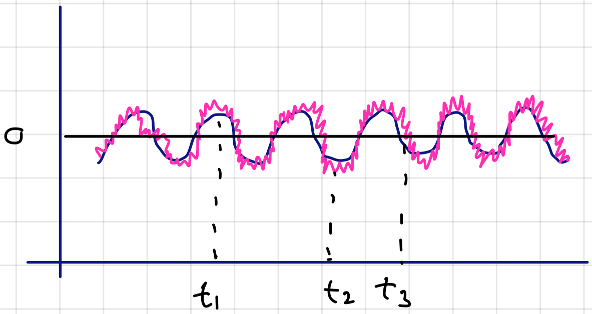
→ These are the required condition for white noise and also distribution can be different at different times, but all should satisfy these three conditions

Remark: WN need not be normally distributed

Ex SNR

$X_t = A\sin(2\pi f t + \phi) + W_t$

→ adding noise



Random variable at t_1, t_2, t_3

Ex Moving average process (MA(1))

$W_t \sim WN(0, \sigma^2_w)$

$X_t = W_t + \theta W_{t-1}$

→ scaling factor

→ It is made from noise

→ linear combination of noise

→ combination of present white noise and earlier white noise

$= 1W_t + \theta W_{t-1} + 0W_{t-2} + 0W_{t-3}$

Autoregressive Process (AR(1))

$W_t \sim WN(0, \sigma^2_w)$

$X_t = \phi X_{t-1} + W_t$

Recursive

$= \phi(\phi X_{t-2} + W_{t-1}) + W_t$

$= W_t + \phi W_{t-1} + \phi^2 X_{t-2}$

$= W_t + \phi W_{t-1} + \phi^2 W_{t-2} + \phi^3 X_{t-3}$

AR(1) process is same as MA(∞) process $|\phi| < 1$

Why is it called autoregressive?

$Y = h(X) + error$

$X_t = \phi X_t + W_t$

→ because it is regression with itself

Brownian Motion

$\{X_t\}$ is said to follow Brownian Motion (BM) if

1. $X_0 = a$ (If $a=0$ Standard BM)

2. $X_{t+s} - X_t$ is independent to X_t → increment does not depend on X_t and amount of jump will depend on time gap/lag

3. $X_{t+s} - X_t \sim N(0, s)$

4. $\{X_t\}$ has a continous path But not differentiable

As much as finer you chop it, you would see jump (notch) making it non-differentiable.
However fine you chop it, there will always be jump

$$W_t \overset{iid}{\sim} N(0,1)$$

$$X_t = \frac{1}{\sqrt{n}} \sum_{k=0}^{[nt]} W_k \quad \Big| \quad \begin{array}{l} n \to \infty \\ t \in (0,1) \end{array}$$

Will have more variance with increasing t

greatest integer function

$$= \frac{\sqrt{[nt]}}{\sqrt{n}} \underbrace{\left(\frac{1}{\sqrt{[nt]}} \sum_{k=1}^{[nt]} W_k \right)}_{N(0,1)} \longrightarrow \sqrt{t}\, N(0,1) \equiv N(0,t)$$

scaling factor, will affect only variance

$$0 < t_1 < t_2 < t_3 \cdots t_k < \cdots$$

→ variance-covariance matrix

$$(X_{t_1}, X_{t_2} \cdots\cdots X_{t_k}) \sim N_k \left(\underset{\sim}{0}, \begin{pmatrix} t_1 & & \min(t_i, t_j) \\ & t_2 & \\ & & \ddots \\ & & & t_k \end{pmatrix} \right)$$

Multivariate normal distribution

If $t < s$

$$\begin{pmatrix} X_t \\ X_s \end{pmatrix} \sim N \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} t & t \\ t & s \end{pmatrix} \right)$$

$n \to$ If you have an interval $(0,1)$, it means how fine you are chopping

$$X_t = \frac{1}{\sqrt{n}} \sum_{k=1}^{[nt]} W_k$$

$$X_s = \frac{1}{\sqrt{n}} \sum_{k=1}^{[ns]} W_k = \boxed{\frac{1}{\sqrt{n}} \left(\sum_{k=1}^{[nt]} W_k \right.} + \left. \sum_{k=[nt]+1}^{[ns]} W_k \right)$$

↳ only this will contribute to covariance as they are iid so after terms won't affect

Brownian Bridge

consider $\{X_t\}$ be a BM

conditioned to $X_0=0$ and $X_t=0$ $\quad t\in[0,1]$

then X_t is said to follow "Standard Brownian Bridge" with the following distributed property

$$B_t \sim \underset{t<s}{B} \text{ Bridge } [0,1]$$

$$\begin{pmatrix} B_t \\ B_s \end{pmatrix} \sim N\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix} \begin{pmatrix} t(1-t) & t(1-s) \\ t(1-s) & s(1-s) \end{pmatrix} \right)$$

$$B(t) = W(t) - t\, W(1) \equiv N(0,1)$$

$W \sim N(0,t)$ or Brownian motion

KS test under H_0 follows

$$K = \max_{0\le t\le 1} |B(t)| \qquad B_t \sim B \text{ Bridge}([0,1])$$

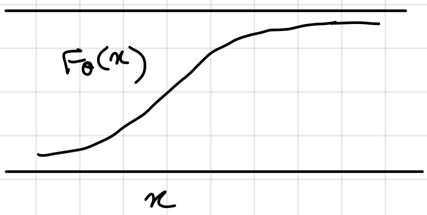
X_1 $\quad X \sim$ iid continuous

H_0 $\quad X \sim F_\theta$

H_1 $\quad X$ does not follow F_θ

If H_0 is true $\Leftrightarrow$ If $X \sim F_\theta(\;\;)$

$T = F_\theta(X) \sim$



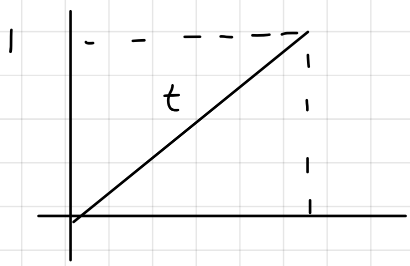
$P(T \le t)$ $\quad t\in[0,1]$

$= P(F_\theta(X) \le t)$

$P(X \le F_\theta^{-1}(t)) = F_\theta(F_\theta^{-1}(t)) = t$

$T = F_\theta(t) \sim U[0,1]$ under H_0 $\rightarrow$ CDF of T will be $U[0,1]$

$T_i = F_\theta(X_i)$



H_0: $\quad T_i \sim U[0,1]$

vs

H_1 $\quad T_i$ does not follow $U[0,1]$

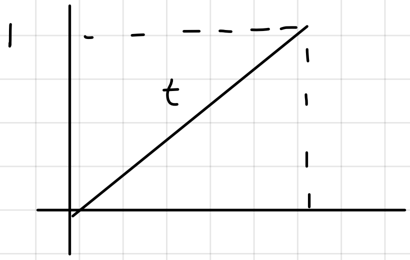
$$G_n(t) = \frac{1}{n}\sum_{i=1}^{n} I_{\{T_i \le t\}}$$

→ indicator function (Dirac Delta function) → each indicator function following bernoulli → sum of bernoulli follows binomial distribution

so $\sum_{i=1}^{nt} I_{T_i \le t}$

$$\sup_t \sqrt{n}\,|G_n(t) - t| \xrightarrow{\text{distribution}} K$$

one is empirical cdf other is true cdf



$$E\left(\frac{1}{n}\sum_{i=1}^{n} I_{\{T_i \le t\}}\right) = \frac{nt}{n} = t$$

$$V\left(\frac{1}{n}\sum_{i=1}^{n} I_{\{T_i \le t\}}\right) = \frac{nt(1-t)}{n^2}$$

$$= \frac{t(1-t)}{n}$$

$$E(G_n(t)) = t$$

$$V(G_n(t)) = \frac{t(1-t)}{n}$$

$$\sqrt{n}\,(G_n(t) - t) \xrightarrow{d} N(0, t(1-t))$$

as $n \to \infty$

it converges in distribution

$$K = \sup_{0<t<1} |B_0(t)|$$